HOPPER INFORMATION SERIES

08 / FIDUCIAL MARKERS

AprilTags with Hopper

Visual IDs for detection, centering and 2D heading alignment.

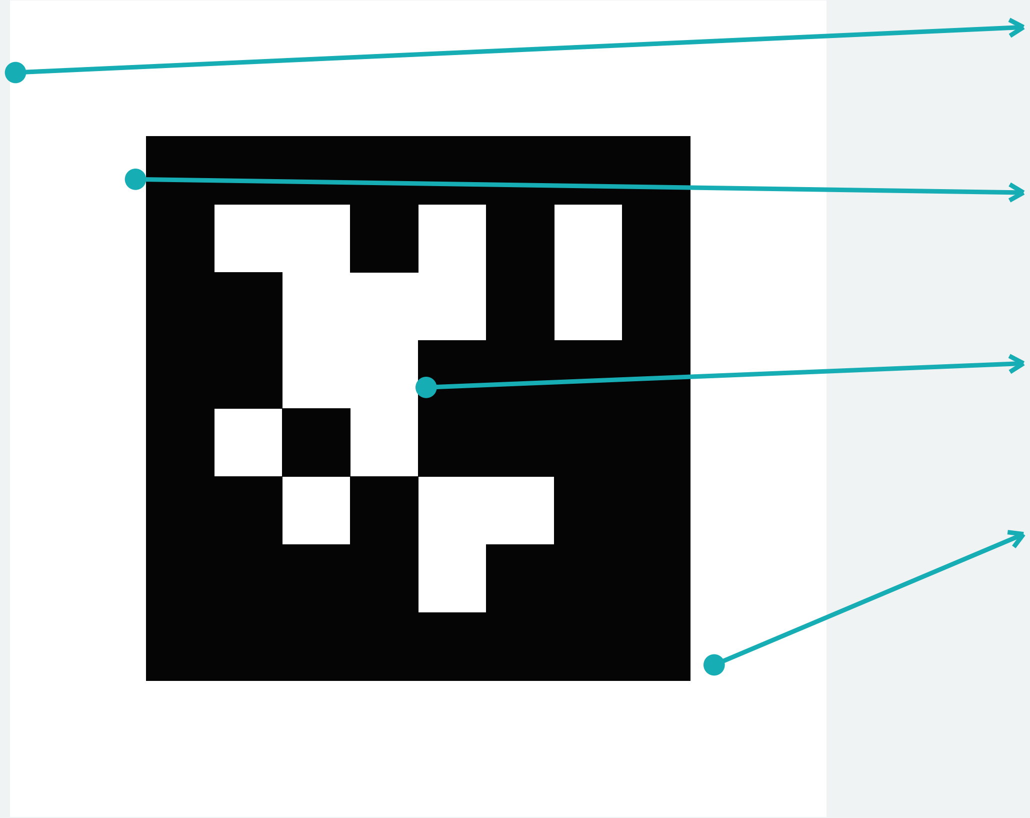
tag36h11

tag36h11 ID 0

ID + orientation

TAG ANATOMY

A visual ID designed for robots



White quiet zone
Keep this margin clear

Black border
Makes a square candidate

Data cells
Encode ID with error separation

Orientation
Pattern defines rotation

tag36h11 ID 0

HOPPER FAMILY

- tag36h11
- IDs 0 through 586
- Black and white only
- Up to 3 payload-bit errors accepted

WHY FIDUCIALS

- Known geometry.
- Unique machine-readable ID.
- Orientation from one image.
- No battery or radio on the tag.

LOCAL TAG36H11 PIPELINE

How Hopper Studio detects a tag

1 Grayscale
Convert RGB camera pixels to luminance.

→

2 Threshold
Estimate a black/white split.

→

3 Squares
Find connected dark square candidates.

→

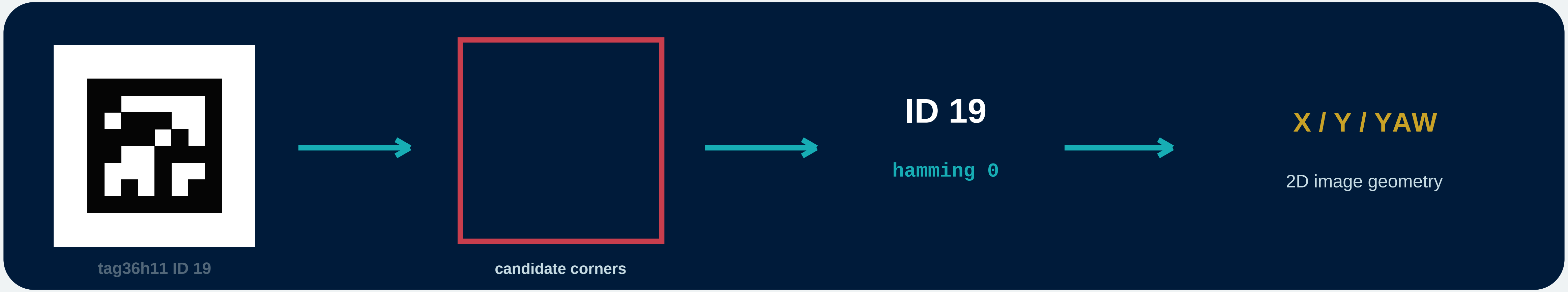
4 Sample
Map each candidate into an 8x8 grid.

→

5 Match
Compare rotations with the codebook.

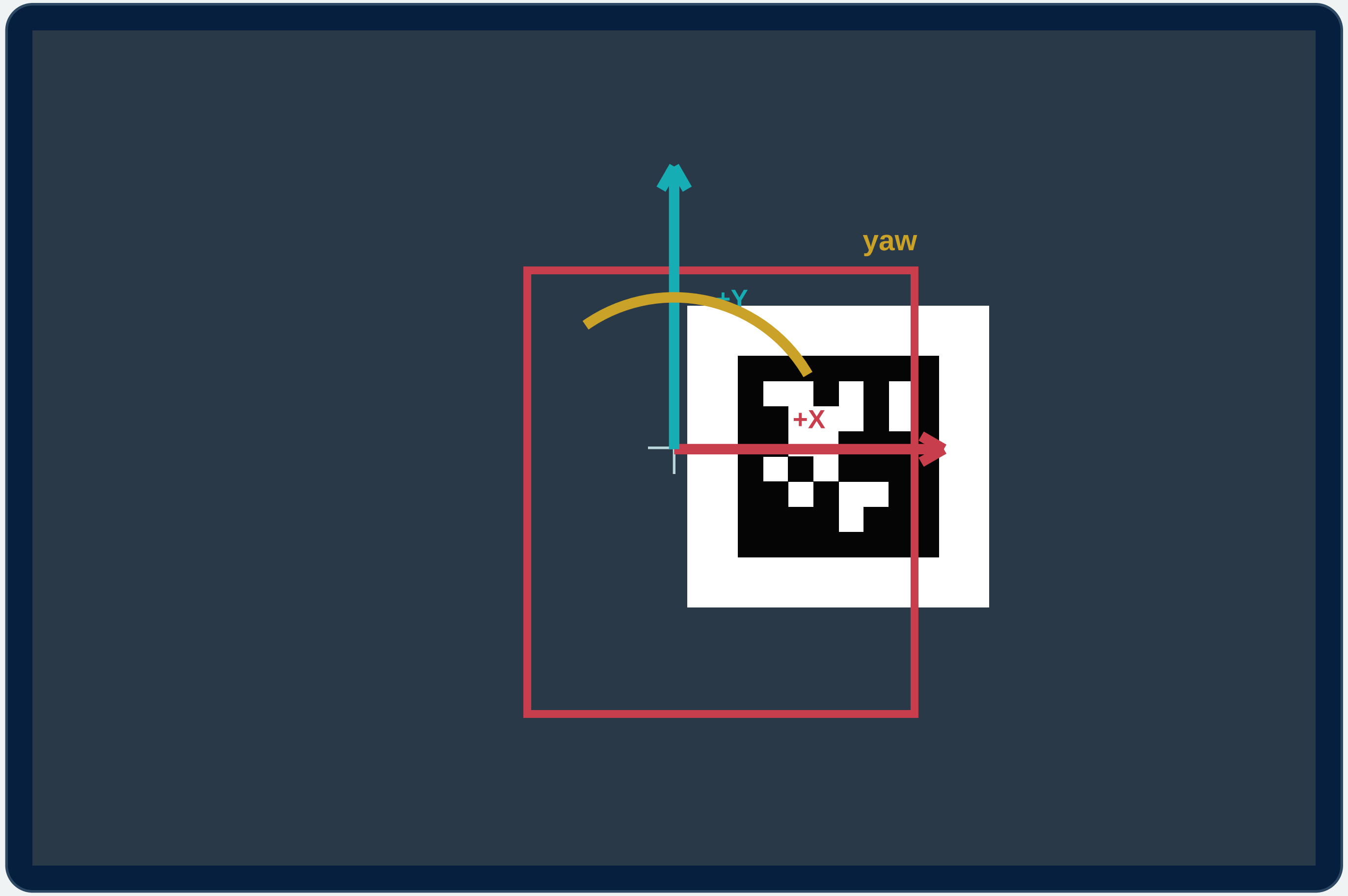
→

6 Pose
Return ID, corners, center and 2D yaw.



WHAT THE RESULT MEANS

Coordinates and pose in Hopper Studio



APRILTAG RESULT

- id + family + hamming
- four corners + center in pixels
- bounding box
- centerX / centerY: -100..100
- yaw: normalized around +/-180 deg

DO NOT OVERCLAIM

- Yaw is 2D image-plane orientation.
- It is not full 3D pose or aircraft heading.
- No distance or altitude is returned.
- centerOnAprilTag never lands.

PHYSICAL TAG SETUP

Print cleanly and place deliberately

tag36h11 ID 7

PRINT IN HOPPER STUDIO

- Vision Testing -> AprilTag Detection.
- Choose an ID from 0-586.
- Generate PDF for a full-page US Letter vector tag.
- Print at 100% scale with a clean white margin.

PLACE FOR THE CAMERA

- Keep the sheet flat and well lit.
- Avoid glare, folds, shadows and clipped margins.
- Start large and close; test before increasing height.
- Keep paper and people clear of propellers.

SCAN, TEST, CENTER

Three block patterns support tag missions

scan for april tags

Fresh scan. Replaces the latest tag list and updates the overlay.

camera sees april tag with ID any / 0-586

Fresh scan. Boolean true when the requested ID is present.

center on april tag [ID] at [power]% | center slack | angle slack | lost searches

1. **Scan** — Choose requested or nearest tag.
2. **Translate** — Correct larger x/y error.
3. **Stabilize** — Reset and wait.
4. **Align** — Timed yaw correction.
5. **Finish** — True when centered + aligned.

Defaults: 10% power | +/-5% center | +/-5 deg angle | 3 lost scans | 30 s hard timeout

MISSION PATTERNS

AprilTags become visual landmarks

CENTERING PAD

Center the camera over one ID, then explicitly land.

ZONE ID

Use different IDs for room regions or task stations.

HEADING CUE

Align the drone's forward axis with the tag x-axis.

SEARCH

Lawnmower scan until any tag or one requested ID appears.

SEQUENCE

Visit IDs in a planned order with a timeout at each.

VERIFY

Require repeated ID/pose agreement before motion.

A tag is a relative visual landmark - not GPS, altitude or collision avoidance.

CHECKLIST AND SOURCES

Reliable tags need reliable experiments

BEFORE FLIGHT

- Confirm the family is tag36h11.
- Test the intended ID, size, angle and lighting.
- Keep the full border and quiet zone in frame.
- Use a lost-tag limit and an overall timeout.
- Plan a safe fallback land.

SOURCES AND FURTHER READING

University of Michigan APRIL Lab - AprilTag
https://april.eecs.umich.edu/software/apriltag

AprilTag 2 paper
https://april.eecs.umich.edu/media/pdfs/wang2016iros.pdf

Hopper Studio detector and printable tag generator
lib/apriltags.ts, lib/vision.ts and README.md

DETECT -> CENTER -> ALIGN -> DECIDE

The mission still owns height, obstacle clearance and landing.